a Answer quality
b Complete model cost
12 graphs per arm, five reader seeds; dashed F1 reference = always-empty baseline (1/6)
0
0.5
1
0k
20k
40k
60k
Qwen3-1.7B
Before
F1 0.771 | EM 0.333
Mean 34,901 tokens
After
F1 0.735 | EM 0.383
Mean 34,622 tokens
Direct
F1 0.691 | EM 0.333
Mean 4,093 tokens
Qwen3-4B
Before
F1 1.000 | EM 1.000
Mean 51,751 tokens
After
F1 1.000 | EM 1.000
Mean 51,612 tokens
Direct
F1 0.697 | EM 0.683
Mean 8,231 tokens
c Paired F1 effects
Existing descriptive 95% intervals
1.7B
4B
Before - after
+0.036 [-0.007, 0.083]
Before - direct
+0.080 [-0.030, 0.180]
Before - after
+0.000 [0.000, 0.000]
Before - direct
+0.303 [0.137, 0.467]